

Suppose Dr. WhyLie comes up to you and claims that he has invented a super-fast comparison-based priority queue. The speed of the priority queue operations are as follows (n is the number of items currently in the priority queue):

- insert a new item in $O(\sqrt{\log n})$ time
- extract (remove and return) the smallest item from the priority queue in $O(\sqrt{\log n})$ time

Dr. WhyLie's algorithm

insert at $O(\sqrt{\log n})$ on n items
extract at $O(\sqrt{\log n})$ on n items

$O(n\sqrt{\log n})$ for $(i \leq n)$
for every item of size input n find the right spot

$O(n\sqrt{\log n})$

$$O(n\sqrt{\log n}) + O(n\sqrt{\log n}) = O(n\sqrt{\log n})$$

Runtime of Dr. WhyLie's algorithm

Dr. WhyLie claims that he has invented a comparison-based priority queue where both insert and extract-min run in $O(\sqrt{\log n})$. If such a priority queue existed, we could use it to sort n items by first inserting all n elements and then extracting the minimum n times. Since each operation takes $O(\sqrt{\log n})$, the total runtime for sorting would be $O(n\sqrt{\log n})$. Based on this, Dr. WhyLie is lying because we know that a comparison-based sorting algorithm requires $\Omega(n \log n)$ to properly ~~fun~~ compare, which grows asymptotically faster than $O(n\sqrt{\log n})$. Therefore, Dr. WhyLie's algorithm cannot exist.

